Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Validation and optimization of measurement tools and material characterization techniques for German and European core industry application			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials		Electrics & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Industrial/general machinery	General machine tool (100605)	Machines using optical technology	Optical measurement*control instruments (200104)
Concept of Proposed Technology	Automated multi-sensor, non-destructive material characterization platform; validates tools, maps current flows/losses for QA in batteries, electrolysis.			
Rationale for Joint R&D Project	Collaboration combines Fraunhofer's characterization expertise with Korean SMEs' automated testing manufacturing to adapt, validate, and transfer multi-sensor tools, delivering TRL5–6 prototypes for factory QA of energy materials.			
Key Words	1. Automated Testing Systems 2. optical, electrical, chemical measuring and calibration device 3. microstructural characterization routines and quality assurance			

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	• Develop an advanced multi-sensor, non-destructive characterization platform: optimized optical/electrical/chemical measurement routines • build automated testing/calibration systems • prototype data processing and microstructure diagnostics • validate TRL5-6 factory-ready tools for spatially resolved mapping of current flows, losses, and properties in energy materials			
R&D Target Specifications	Target TRL 6 (start TRL 3); approved setup TRL 5/6 for factory deployment. Non-destructive, spatially resolved localization of current flows (resolution 1 cm for measurement distances below 5 cm), loss mechanisms (resolution 1 mm for loss-sources within 1 mm below the outer surface), material properties (resolution ~50 µm for elemental composition); detects performance limitations and production failures; enables battery cell mapping on areas of up to 20 x 20 cm and transfer to other industrial applications.			

R&D Plan

Phase	TRL goal	Research approach & methodologies	Key tasks	Expected milestones
1	3 (lab-scale verification)	Multi-sensor characterization; optical/electrical/chemical measurement; microstructural diagnostics	Transfer characterization tools from abroad to meet German/EU market demands; prototype data routines	Basic performance verified; initial data routines established
2	4 (key performance evaluation)	Optimization of multi-sensor testing procedures; QA-oriented microstructure analysis	Optimize testing procedures; develop measuring/calibration device integration	Key performance evaluated; optimized procedures; validated characterization
3	5 (prototype production/ evaluation)	Advanced testing systems; automated testing solutions	Develop prototypes of automated testing systems; integrate multi-sensor tool	Prototype non-destructive measurement tool for spatial localization of current flows, losses, material properties
4	6 (pilot-scale validation)	Factory implementation; application transfer to industry (battery, electrolysis)	Deploy approved measurement setup in Korean factory environment; enable detection of performance limits and production failures	TRL-5/6 approved setup; battery cell mappings informing improved design; transfer to other industrial applications

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Know-how provider and technology transfer to different industrial applications like battery and energy material manufacturers • development/optimization of multi-sensor tool for new characterization measurement routines • working as material characterization platform for beneficial enabling of industry demand • Understanding the production process and the optimal material design to identify relevant material characteristics • Using cross-scale microstructure data and diagnostics to support the optimization of relevant process parameters
Expected R&R of Korean SME Partner	• specialized in the development and manufacturing of testing systems applicable across various industries. • offering automated testing systems, specialized test stands, measuring and calibration devices • leading role due to its extensive expertise in developing advanced testing systems and automated testing solutions • collaboration with other partners, implementing their know-how and focus on quality assurance enhancing energy material, f.e. and battery as well as electrolysis performance

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	• new non-destructive measurement tool for spatially resolved localization of current flows, loss mechanisms and material properties of energy materials • approved measurement setup (TRL-5/6) for the implementation into several characterization requests in the factory environment in Korea, enabling the detection and significantly improved understanding of performance limitations and production failures. • example battery: improved design based on the insights gained from battery cell mappings. By applying this new methodology on a larger scale, Korean battery manufacturers can significantly enhance their products and production technologies, thereby better competing with international competitors.

6. References

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(2)

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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